

# Wave 1 Statistical Analyses

Marta

## Hierarchical Bayesian Models Wave 1

This document runs the models used to analyse the main outcomes for Wave 1. The structure of the document follows the Analysis\_Plans.txt file. The variables in the datasets are explained in Meta-Data.txt.

### Data Set 1 - Binary Categorical Outcomes

#### Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).  
The explainer variables are: \* Whether the participant was at a university that was affiliated with the UKRN  
\* Whether the participant was in their first, second, placement or third year  
\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean  sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.10    0.01 0.44   -0.77   -0.20    0.10    0.39    0.95  2825
## sigma        1.12    0.00 0.24    0.72    0.95    1.09    1.26    1.68 10834
## alpha[1]     -0.81    0.01 0.63   -2.08   -1.22   -0.81   -0.40    0.41  3038
## alpha[2]      0.89    0.01 0.76   -0.56    0.38    0.88    1.39    2.46  4156
## alpha[3]      0.26    0.01 0.69   -1.10   -0.19    0.26    0.71    1.62  3627
## alpha[4]      0.26    0.01 0.72   -1.16   -0.22    0.26    0.74    1.70  3988
## alpha[5]     -1.39    0.01 0.75   -2.91   -1.87   -1.36   -0.88    0.02  4211
## alpha[6]      0.86    0.01 0.80   -0.67    0.32    0.84    1.38    2.49  4830
## alpha[7]     -1.22    0.01 0.75   -2.73   -1.71   -1.21   -0.72    0.23  4357
## alpha[8]      1.87    0.01 0.82    0.38    1.31    1.83    2.38    3.60  5153
## alpha[9]     -0.09    0.01 0.59   -1.23   -0.48   -0.09    0.30    1.07  6555
## alpha[10]     2.01    0.01 0.87    0.51    1.41    1.94    2.54    3.92  8424
## alpha[11]     0.54    0.01 0.69   -0.80    0.07    0.53    1.00    1.93  6671
## alpha[12]    -1.45    0.01 0.76   -3.04   -1.92   -1.41   -0.93   -0.07  8816
## alpha[13]     0.38    0.01 0.53   -0.66    0.02    0.37    0.73    1.41  6785
## alpha[14]    -0.01    0.01 0.53   -1.04   -0.37   -0.01    0.34    1.05  6341
## alpha[15]    -1.26    0.01 0.69   -2.70   -1.69   -1.23   -0.79    0.02  7325
## alpha[16]     0.97    0.01 0.45    0.12    0.66    0.96    1.27    1.86  5231
## alpha[17]    -0.16    0.01 0.49   -1.14   -0.49   -0.16    0.17    0.79  6866
## alpha[18]    -1.18    0.01 0.47   -2.15   -1.49   -1.17   -0.86   -0.29  6560
## alpha[19]     0.28    0.01 0.57   -0.81   -0.12    0.27    0.66    1.40  5938
## alpha[20]     0.54    0.01 0.40   -0.25    0.26    0.53    0.81    1.35  4086
## alpha[21]     0.37    0.01 0.41   -0.43    0.10    0.38    0.65    1.19  5155
## alpha[22]     0.41    0.01 0.41   -0.38    0.14    0.41    0.69    1.22  4130
## alpha[23]     0.11    0.01 0.67   -1.19   -0.34    0.10    0.55    1.42  7922
```

```

## beta1      -0.38    0.01 0.69   -1.72   -0.83   -0.38    0.08    0.99   3420
## beta2       0.07    0.00 0.15   -0.22   -0.03    0.07    0.17    0.36   2743
## beta3       0.81    0.00 0.24    0.34    0.65    0.81    0.97    1.28   5594
## lp__      -477.25    0.05 4.12 -486.26 -479.80 -476.90 -474.33 -470.18  6342
##           Rhat
## mu          1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1
## alpha[10]    1
## alpha[11]    1
## alpha[12]    1
## alpha[13]    1
## alpha[14]    1
## alpha[15]    1
## alpha[16]    1
## alpha[17]    1
## alpha[18]    1
## alpha[19]    1
## alpha[20]    1
## alpha[21]    1
## alpha[22]    1
## alpha[23]    1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Thu Feb 19 23:50:16 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

#### Number of participants in the analysis

Total N = 803

UKRN N = 421

Non-UKRN N = 382

Year 1 N = 326

Year 2 N = 261

Placement Year N = 57

Year 4 N = 159

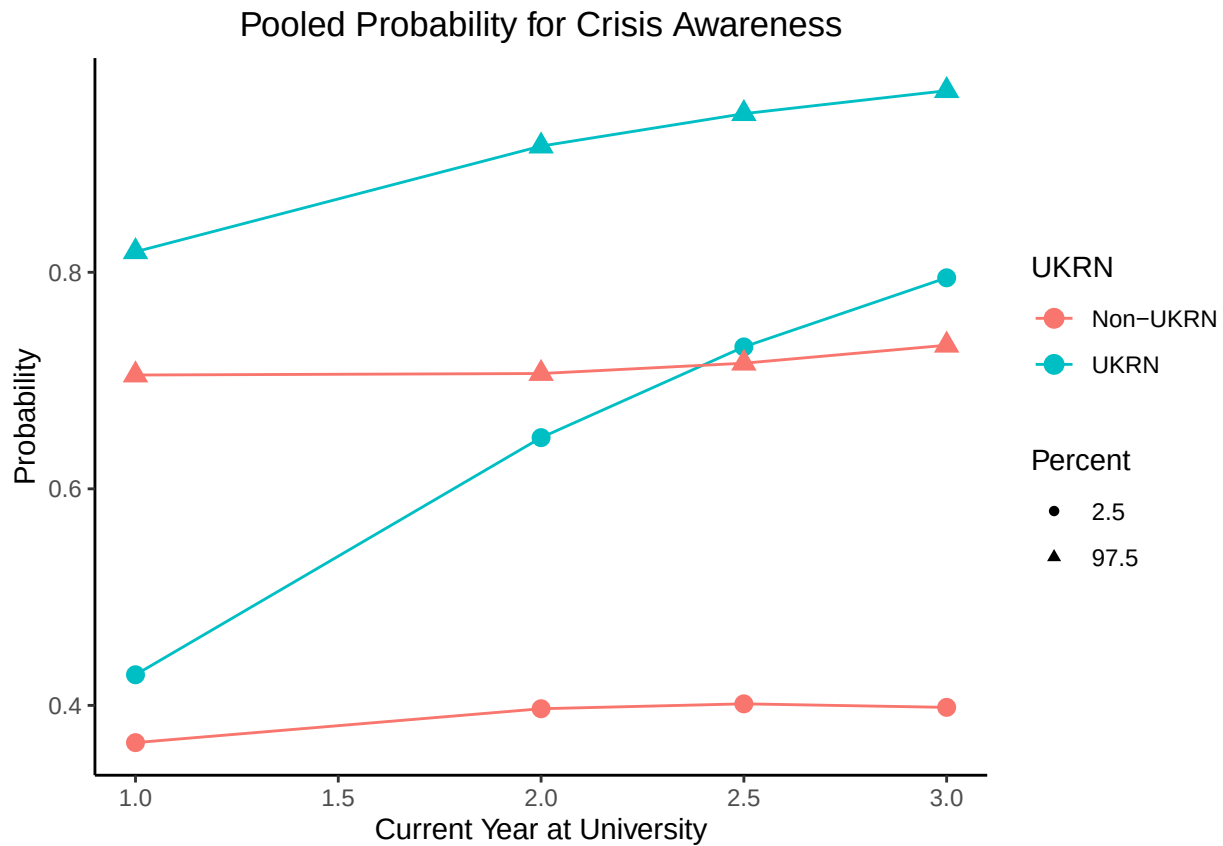
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.3388522

97.5% = 1.2832707

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the reproducibility crisis in the later years of education. For students who attend a non-UKRN affiliated

university, there is no difference.



### Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).

The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement of third year

\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           -2.10    0.01 0.49   -3.08   -2.43   -2.09   -1.77   -1.17  1900
## sigma         0.77    0.00 0.21    0.42    0.62    0.74    0.89    1.26  5305
## alpha[1]      -2.68    0.01 0.60   -3.86   -3.08   -2.67   -2.27   -1.50  2200
## alpha[2]      -1.69    0.01 0.66   -2.98   -2.14   -1.69   -1.25   -0.39  2584
## alpha[3]      -2.26    0.01 0.64   -3.52   -2.68   -2.25   -1.83   -1.01  2498
## alpha[4]      -1.64    0.01 0.66   -2.93   -2.08   -1.64   -1.19   -0.35  2639
## alpha[5]      -2.96    0.01 0.78   -4.60   -3.45   -2.92   -2.43   -1.51  3352
## alpha[6]      -1.44    0.01 0.68   -2.75   -1.89   -1.44   -0.99   -0.10  2715
## alpha[7]      -3.08    0.01 0.75   -4.65   -3.56   -3.05   -2.56   -1.69  3119
## alpha[8]      -1.12    0.01 0.63   -2.35   -1.54   -1.12   -0.70    0.14  2398
## alpha[9]      -1.62    0.01 0.66   -2.94   -2.06   -1.61   -1.17   -0.35  3533
## alpha[10]     -1.57    0.01 0.68   -2.91   -2.03   -1.57   -1.12   -0.26  3625
```

```

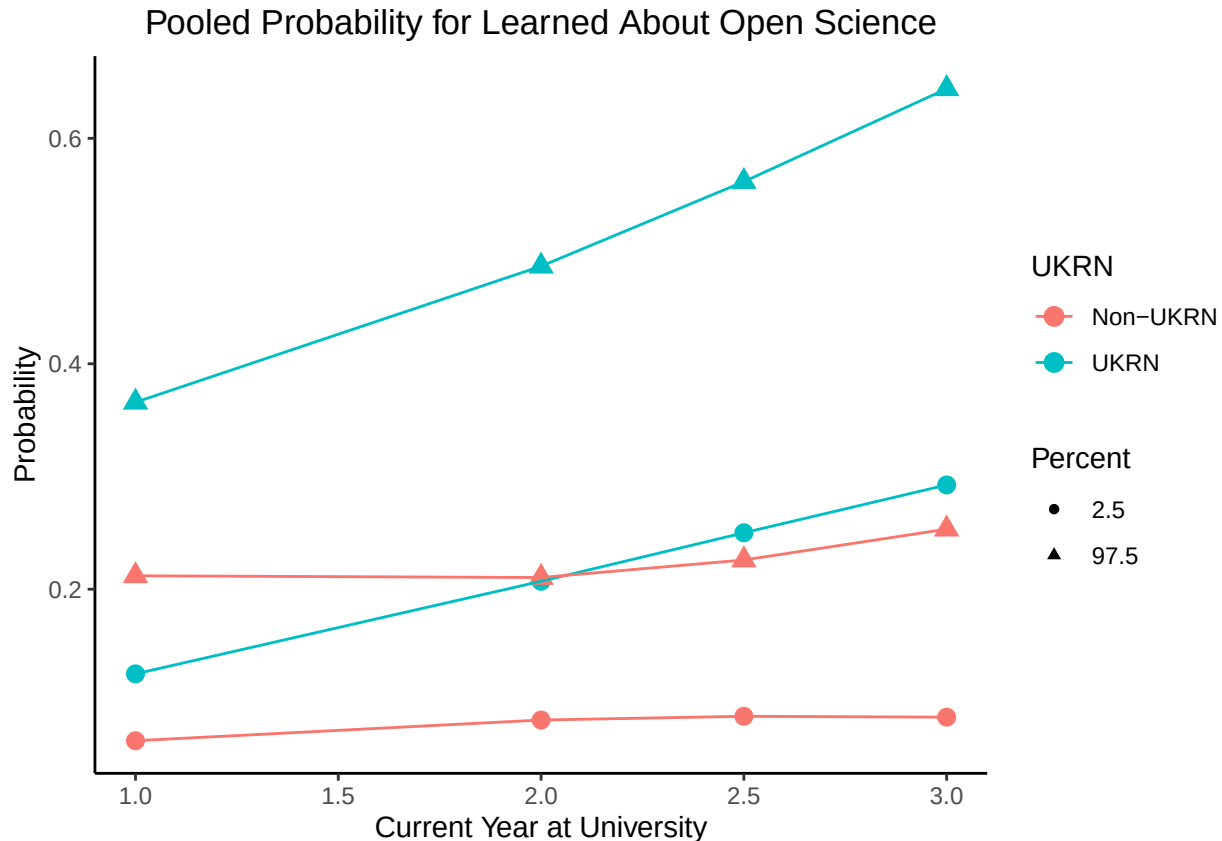
## alpha[11] -2.71 0.01 0.85 -4.51 -3.23 -2.65 -2.13 -1.18 3818
## alpha[12] -2.66 0.01 0.81 -4.40 -3.17 -2.61 -2.10 -1.19 4474
## alpha[13] -2.02 0.01 0.64 -3.33 -2.44 -2.01 -1.58 -0.80 3904
## alpha[14] -2.29 0.01 0.68 -3.69 -2.73 -2.27 -1.82 -0.99 3663
## alpha[15] -2.06 0.01 0.70 -3.49 -2.52 -2.05 -1.58 -0.73 3776
## alpha[16] -2.26 0.01 0.57 -3.40 -2.64 -2.24 -1.87 -1.19 3253
## alpha[17] -1.68 0.01 0.58 -2.84 -2.07 -1.67 -1.28 -0.57 3601
## alpha[18] -3.13 0.01 0.77 -4.85 -3.60 -3.07 -2.59 -1.78 4321
## alpha[19] -1.82 0.01 0.66 -3.15 -2.26 -1.82 -1.37 -0.54 3305
## alpha[20] -1.79 0.01 0.52 -2.82 -2.13 -1.78 -1.43 -0.80 2543
## alpha[21] -1.53 0.01 0.51 -2.56 -1.87 -1.53 -1.18 -0.54 2944
## alpha[22] -2.54 0.01 0.57 -3.72 -2.91 -2.53 -2.14 -1.45 3016
## alpha[23] -1.84 0.01 0.73 -3.29 -2.32 -1.83 -1.35 -0.42 3829
## beta1 0.32 0.01 0.66 -0.97 -0.12 0.32 0.76 1.61 2429
## beta2 0.14 0.00 0.20 -0.25 0.00 0.14 0.27 0.53 1991
## beta3 0.41 0.00 0.26 -0.09 0.24 0.41 0.59 0.92 2932
## lp__ -384.73 0.07 4.78 -394.89 -387.77 -384.44 -381.41 -376.08 4283
## Rhat
## mu 1
## sigma 1
## alpha[1] 1
## alpha[2] 1
## alpha[3] 1
## alpha[4] 1
## alpha[5] 1
## alpha[6] 1
## alpha[7] 1
## alpha[8] 1
## alpha[9] 1
## alpha[10] 1
## alpha[11] 1
## alpha[12] 1
## alpha[13] 1
## alpha[14] 1
## alpha[15] 1
## alpha[16] 1
## alpha[17] 1
## alpha[18] 1
## alpha[19] 1
## alpha[20] 1
## alpha[21] 1
## alpha[22] 1
## alpha[23] 1
## beta1 1
## beta2 1
## beta3 1
## lp__ 1
##
## Samples were drawn using NUTS(diag_e) at Thu Feb 19 23:54:03 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 798  
 UKRN N = 420  
 Non-UKRN N = 378  
 Year 1 N = 324  
 Year 2 N = 258  
 Placement Year N = 57  
 Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. The results are non-significant.



### Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement of third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5%  n_eff
## mu          -1.37     0.01  0.37   -2.11   -1.62   -1.37   -1.12   -0.66  1214
## sigma        0.37     0.00  0.14    0.14    0.27    0.36    0.46    0.69  1819
## alpha[1]     -1.69     0.01  0.44   -2.59   -1.98   -1.68   -1.40   -0.86  1467
## alpha[2]     -1.48     0.01  0.48   -2.46   -1.79   -1.46   -1.15   -0.57  1795
```

```

## alpha[3]      -1.46      0.01 0.46   -2.40   -1.77   -1.45   -1.15   -0.59  1642
## alpha[4]      -1.17      0.01 0.47   -2.07   -1.48   -1.17   -0.86   -0.24  1804
## alpha[5]      -1.58      0.01 0.51   -2.62   -1.90   -1.55   -1.23   -0.64  1849
## alpha[6]      -1.09      0.01 0.48   -2.01   -1.42   -1.11   -0.77   -0.13  1852
## alpha[7]      -1.51      0.01 0.49   -2.51   -1.83   -1.50   -1.18   -0.59  1857
## alpha[8]      -0.98      0.01 0.46   -1.87   -1.29   -0.98   -0.67   -0.06  1626
## alpha[9]      -1.29      0.01 0.48   -2.24   -1.60   -1.29   -0.98   -0.36  1938
## alpha[10]     -1.37      0.01 0.48   -2.35   -1.69   -1.37   -1.05   -0.44  1972
## alpha[11]     -1.42      0.01 0.51   -2.47   -1.73   -1.40   -1.08   -0.44  1995
## alpha[12]     -1.64      0.01 0.53   -2.81   -1.97   -1.61   -1.28   -0.69  2054
## alpha[13]     -1.48      0.01 0.47   -2.44   -1.78   -1.46   -1.16   -0.58  2030
## alpha[14]     -1.48      0.01 0.48   -2.46   -1.79   -1.47   -1.17   -0.58  1927
## alpha[15]     -0.93      0.01 0.52   -1.90   -1.29   -0.96   -0.60    0.13  1998
## alpha[16]     -1.54      0.01 0.43   -2.42   -1.83   -1.52   -1.25   -0.72  1790
## alpha[17]     -1.44      0.01 0.45   -2.35   -1.73   -1.43   -1.13   -0.59  2008
## alpha[18]     -1.43      0.01 0.43   -2.28   -1.71   -1.42   -1.14   -0.61  1918
## alpha[19]     -1.59      0.01 0.51   -2.64   -1.90   -1.56   -1.24   -0.65  1934
## alpha[20]     -1.05      0.01 0.41   -1.84   -1.33   -1.05   -0.77   -0.25  1626
## alpha[21]     -1.18      0.01 0.41   -1.98   -1.45   -1.18   -0.90   -0.36  1841
## alpha[22]     -1.33      0.01 0.41   -2.16   -1.60   -1.33   -1.05   -0.54  1680
## alpha[23]     -1.35      0.01 0.50   -2.34   -1.68   -1.35   -1.02   -0.39  2030
## beta1         -0.02      0.01 0.51   -1.01   -0.37   -0.03    0.32    1.00  1775
## beta2          0.07      0.00 0.16   -0.25   -0.04    0.07    0.18    0.38  1358
## beta3          0.27      0.00 0.22   -0.17    0.12    0.27    0.42    0.71  2070
## lp__          -438.62    0.20 7.38 -452.12 -443.60 -439.01 -434.19 -422.33  1345
##
## Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## alpha[20]     1
## alpha[21]     1
## alpha[22]     1
## alpha[23]     1
## beta1         1
## beta2         1
## beta3         1

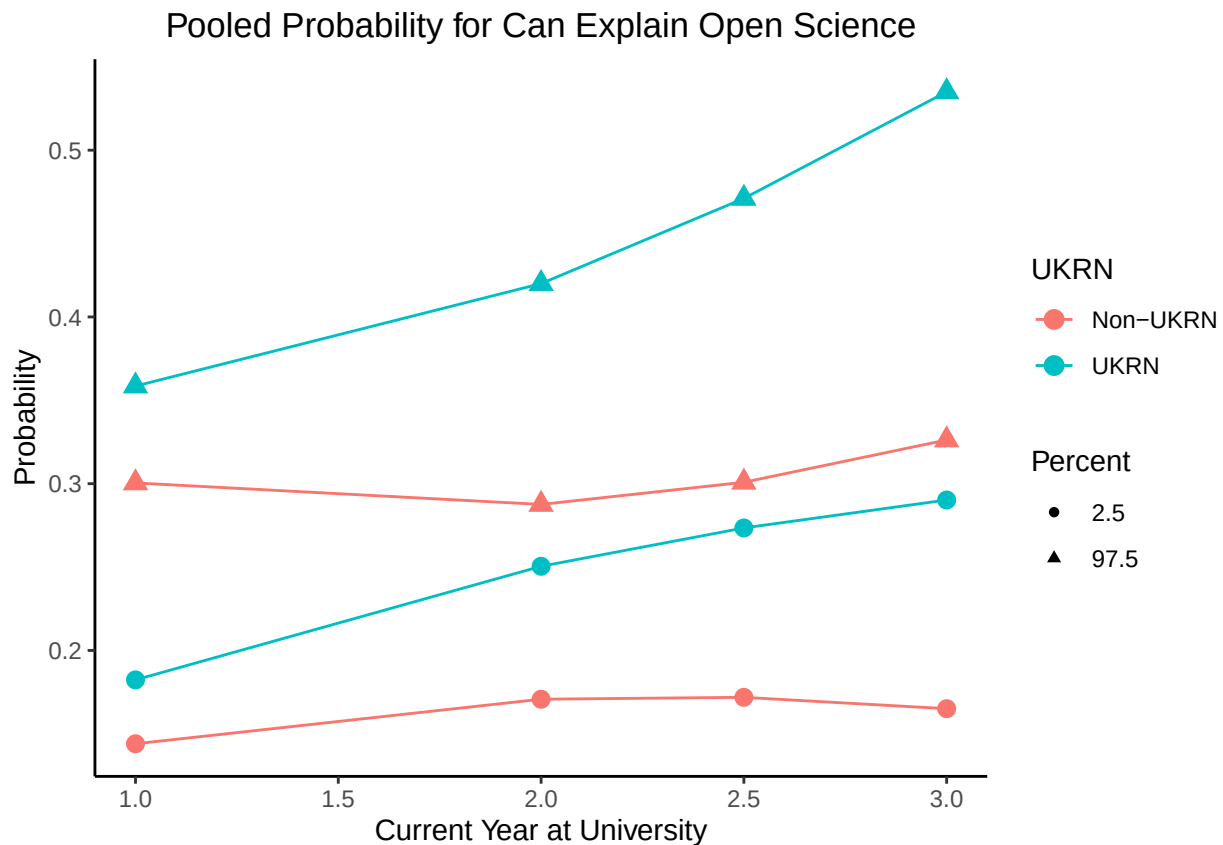
```

```
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Thu Feb 19 23:56:46 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

### Number of participants in the analysis

Total N = 798  
 UKRN N = 420  
 Non-UKRN N = 378  
 Year 1 N = 324  
 Year 2 N = 258  
 Placement Year N = 57  
 Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. No significant results.



### Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no).

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement of third year
- \* Interaction between the two variables above

```

## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##      mean se_mean  sd  2.5%  25%  50%  75%  97.5% n_eff
## mu      3.30    0.02 0.63   2.09   2.87   3.29   3.72   4.59 1586
## sigma    0.84    0.00 0.27   0.37   0.65   0.82   1.00   1.44 3091
## alpha[1] 3.01    0.02 0.76   1.52   2.50   3.00   3.51   4.52 1848
## alpha[2] 2.61    0.02 0.84   0.95   2.05   2.62   3.18   4.25 2164
## alpha[3] 3.49    0.02 0.84   1.91   2.93   3.47   4.03   5.17 2308
## alpha[4] 3.74    0.02 0.91   2.04   3.11   3.69   4.31   5.61 2597
## alpha[5] 2.55    0.02 0.85   0.87   1.99   2.55   3.12   4.21 2258
## alpha[6] 3.97    0.02 1.01   2.16   3.28   3.90   4.59   6.13 2781
## alpha[7] 3.39    0.02 0.93   1.63   2.76   3.36   3.99   5.26 2709
## alpha[8] 3.69    0.02 0.86   2.04   3.11   3.66   4.25   5.47 2432
## alpha[9] 3.39    0.02 0.88   1.75   2.80   3.36   3.96   5.24 3051
## alpha[10] 3.37    0.02 0.87   1.72   2.78   3.34   3.93   5.17 2999
## alpha[11] 3.32    0.02 0.91   1.61   2.70   3.30   3.91   5.15 2925
## alpha[12] 3.73    0.02 0.98   1.97   3.05   3.66   4.34   5.84 3457
## alpha[13] 2.05    0.02 0.76   0.63   1.53   2.02   2.56   3.60 2498
## alpha[14] 3.51    0.02 0.87   1.89   2.92   3.46   4.06   5.32 2894
## alpha[15] 3.33    0.02 0.88   1.68   2.73   3.30   3.89   5.14 3036
## alpha[16] 3.89    0.02 0.81   2.40   3.33   3.85   4.40   5.62 2778
## alpha[17] 3.52    0.02 0.85   1.99   2.93   3.48   4.05   5.33 3111
## alpha[18] 3.07    0.01 0.72   1.71   2.58   3.05   3.54   4.53 2770
## alpha[19] 2.62    0.02 0.80   1.08   2.08   2.62   3.15   4.22 2422
## alpha[20] 4.21    0.02 0.84   2.72   3.63   4.16   4.73   6.00 2499
## alpha[21] 4.22    0.02 0.92   2.62   3.59   4.15   4.78   6.22 3006
## alpha[22] 2.56    0.01 0.66   1.30   2.10   2.55   2.98   3.89 2043
## alpha[23] 2.73    0.02 0.89   1.00   2.14   2.73   3.32   4.49 3111
## beta1     -1.67    0.02 0.86  -3.36  -2.24  -1.67  -1.10   0.01 2142
## beta2     -0.36    0.01 0.25  -0.86  -0.53  -0.36  -0.19   0.13 1670
## beta3      0.88    0.01 0.38   0.16   0.63   0.87   1.13   1.63 3018
## lp__     -240.27    0.12 5.86 -251.84 -244.08 -240.26 -236.55 -228.53 2550
##      Rhat
## mu      1
## sigma    1
## alpha[1] 1
## alpha[2] 1
## alpha[3] 1
## alpha[4] 1
## alpha[5] 1
## alpha[6] 1
## alpha[7] 1
## alpha[8] 1
## alpha[9] 1
## alpha[10] 1
## alpha[11] 1
## alpha[12] 1
## alpha[13] 1
## alpha[14] 1
## alpha[15] 1
## alpha[16] 1
## alpha[17] 1

```

```

## alpha[18]      1
## alpha[19]      1
## alpha[20]      1
## alpha[21]      1
## alpha[22]      1
## alpha[23]      1
## beta1          1
## beta2          1
## beta3          1
## lp__           1
##
## Samples were drawn using NUTS(diag_e) at Thu Feb 19 23:59:57 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

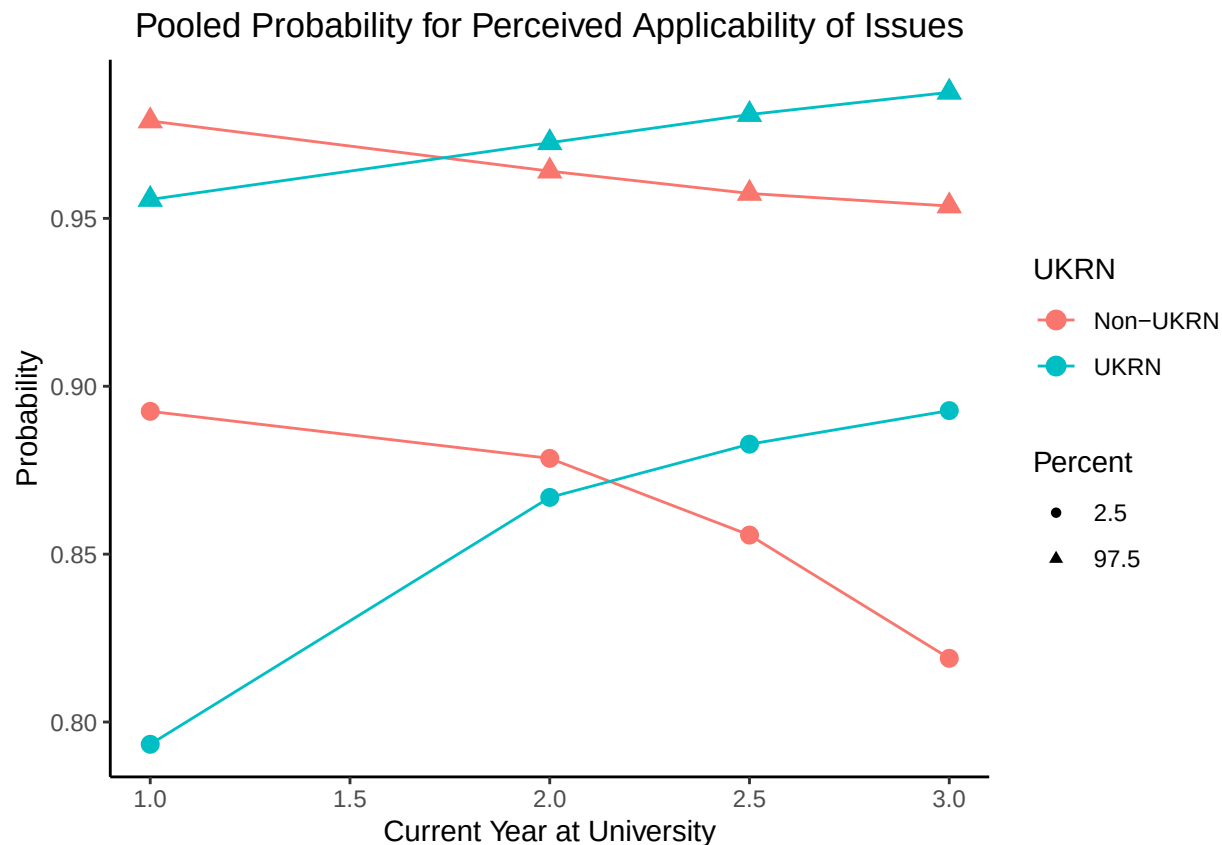
#### Number of participants in the analysis

**Total N** = 798  
**UKRN N** = 420  
**Non-UKRN N** = 378  
**Year 1 N** = 324  
**Year 2 N** = 258  
**Placement Year N** = 57  
**Year 4 N** = 159

The output shows that beta credibility intervals do not cross 0 the interaction term.

2.5% = 0.1566116  
 97.5% = 1.6328406

The interpretation is that students who attend a UKRN affiliated university are more likely to think that the issues around open and reproducible research are applicable to their work in the later years of education.



## Data Set 2 - Truncated Poisson Regression

### Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8. The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           1.54    0.00 0.08    1.37    1.48    1.54    1.59    1.70  1744
## sigma        0.12    0.00 0.04    0.04    0.09    0.11    0.14    0.20  1271
## alpha[1]     1.51    0.00 0.10    1.31    1.45    1.51    1.58    1.72  2073
## alpha[2]     1.54    0.00 0.12    1.31    1.46    1.54    1.61    1.77  2795
## alpha[3]     1.51    0.00 0.11    1.29    1.44    1.52    1.59    1.74  2411
## alpha[4]     1.62    0.00 0.12    1.39    1.54    1.62    1.70    1.87  2706
## alpha[5]     1.43    0.00 0.12    1.17    1.35    1.44    1.52    1.66  2562
## alpha[6]     1.61    0.00 0.12    1.37    1.53    1.60    1.68    1.86  2863
## alpha[7]     1.50    0.00 0.12    1.26    1.42    1.51    1.58    1.74  2869
## alpha[8]     1.57    0.00 0.11    1.35    1.49    1.57    1.64    1.79  2470
## alpha[9]     1.51    0.00 0.12    1.28    1.44    1.52    1.59    1.74  3246
## alpha[10]    1.57    0.00 0.12    1.34    1.49    1.57    1.65    1.81  3395
```

```

## alpha[11]    1.57    0.00 0.13    1.32    1.49    1.57    1.65    1.83    3262
## alpha[12]    1.58    0.00 0.12    1.35    1.50    1.58    1.66    1.84    3490
## alpha[13]    1.46    0.00 0.11    1.22    1.38    1.46    1.53    1.67    2912
## alpha[14]    1.43    0.00 0.11    1.19    1.35    1.43    1.51    1.64    2694
## alpha[15]    1.63    0.00 0.13    1.39    1.54    1.62    1.71    1.88    3312
## alpha[16]    1.41    0.00 0.10    1.20    1.34    1.41    1.48    1.60    2213
## alpha[17]    1.38    0.00 0.12    1.14    1.30    1.39    1.47    1.60    2507
## alpha[18]    1.66    0.00 0.10    1.47    1.59    1.66    1.73    1.86    2967
## alpha[19]    1.52    0.00 0.11    1.29    1.45    1.52    1.60    1.74    2966
## alpha[20]    1.62    0.00 0.09    1.43    1.55    1.61    1.68    1.80    2528
## alpha[21]    1.57    0.00 0.09    1.38    1.51    1.57    1.63    1.75    2903
## alpha[22]    1.63    0.00 0.10    1.44    1.56    1.62    1.69    1.82    2552
## alpha[23]    1.51    0.00 0.12    1.27    1.44    1.51    1.59    1.75    3506
## beta1        -0.08    0.00 0.12   -0.31   -0.16   -0.08    0.00    0.16    2563
## beta2        -0.04    0.00 0.04   -0.11   -0.06   -0.04   -0.01    0.03    1879
## beta3         0.12    0.00 0.05    0.02    0.09    0.12    0.16    0.23    3217
## lp__         1892.54    0.28 6.23 1881.18 1888.67 1892.27 1895.93 1906.23    495
##              Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## alpha[20]     1
## alpha[21]     1
## alpha[22]     1
## alpha[23]     1
## beta1         1
## beta2         1
## beta3         1
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 00:16:16 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

**Total N** = 842  
**UKRN N** = 435  
**Non-UKRN N** = 407  
**Year 1 N** = 344  
**Year 2 N** = 273  
**Placement Year N** = 59  
**Year 4 N** = 166

| Var1 | Freq |
|------|------|
| 0    | 34   |
| 1    | 53   |
| 2    | 81   |
| 3    | 111  |
| 4    | 150  |
| 5    | 146  |
| 6    | 111  |
| 7    | 77   |
| 8    | 79   |

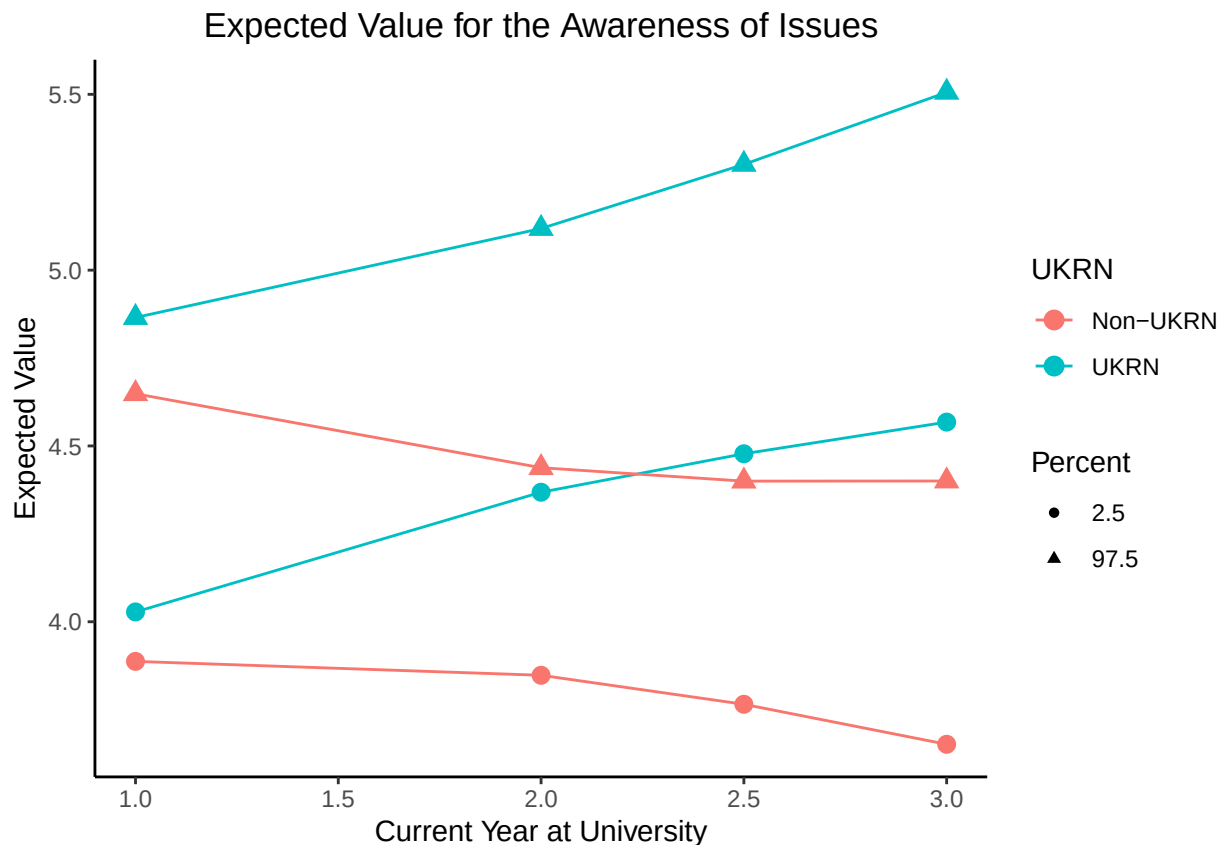
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0229019

97.5% = 0.225566

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the issues around open and reproducible research in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

117 divergent transitions.



### Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- \* Whether the participant was at a university that was affiliated with the UKRN
- \* Whether the participant was in their first, second, placement or third year
- \* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.78    0.03  0.10    0.58    0.71    0.78    0.87    0.93     9
## sigma        0.06    0.01  0.04    0.01    0.02    0.05    0.08    0.15     8
## alpha[1]     0.80    0.03  0.10    0.59    0.72    0.81    0.88    0.97    17
## alpha[2]     0.77    0.04  0.12    0.52    0.69    0.77    0.88    0.94     9
## alpha[3]     0.78    0.03  0.11    0.55    0.70    0.78    0.88    0.95    11
## alpha[4]     0.79    0.03  0.11    0.57    0.72    0.81    0.88    0.98    17
## alpha[5]     0.75    0.04  0.12    0.48    0.67    0.75    0.87    0.93     9
## alpha[6]     0.78    0.03  0.11    0.54    0.70    0.79    0.88    0.97    13
## alpha[7]     0.74    0.05  0.13    0.45    0.65    0.74    0.86    0.92     8
## alpha[8]     0.83    0.01  0.11    0.62    0.76    0.85    0.89    1.04   153
## alpha[9]     0.75    0.04  0.13    0.47    0.66    0.75    0.87    0.93     8
## alpha[10]    0.74    0.05  0.13    0.44    0.65    0.74    0.86    0.92     8
```

```

## alpha[11]    0.77    0.04  0.12    0.51    0.69    0.77    0.87    0.95    11
## alpha[12]    0.80    0.03  0.11    0.57    0.72    0.81    0.89    0.99    17
## alpha[13]    0.78    0.03  0.11    0.55    0.70    0.78    0.88    0.95    11
## alpha[14]    0.77    0.03  0.11    0.54    0.69    0.78    0.88    0.95    11
## alpha[15]    0.79    0.03  0.11    0.55    0.71    0.80    0.88    0.97    15
## alpha[16]    0.81    0.02  0.10    0.60    0.73    0.82    0.89    0.99    20
## alpha[17]    0.79    0.03  0.11    0.57    0.71    0.80    0.88    0.96    13
## alpha[18]    0.78    0.03  0.11    0.56    0.70    0.78    0.88    0.94    10
## alpha[19]    0.78    0.03  0.11    0.54    0.70    0.79    0.88    0.96    11
## alpha[20]    0.78    0.03  0.10    0.57    0.71    0.79    0.87    0.94    10
## alpha[21]    0.79    0.03  0.10    0.58    0.72    0.80    0.89    0.96    12
## alpha[22]    0.76    0.04  0.12    0.53    0.68    0.76    0.88    0.92     7
## alpha[23]    0.83    0.01  0.11    0.61    0.75    0.84    0.89    1.05   128
## beta1        -0.16    0.04  0.13   -0.37   -0.28   -0.17   -0.07    0.11    10
## beta2         0.05    0.02  0.05   -0.02    0.00    0.05    0.08    0.14     7
## beta3         0.09    0.03  0.07   -0.04    0.04    0.08    0.15    0.18     6
## lp__         -207.85    7.57 15.79 -232.29 -220.13 -211.10 -196.12 -177.34     4
##              Rhat
## mu           1.19
## sigma        1.18
## alpha[1]     1.11
## alpha[2]     1.17
## alpha[3]     1.15
## alpha[4]     1.10
## alpha[5]     1.18
## alpha[6]     1.13
## alpha[7]     1.20
## alpha[8]     1.04
## alpha[9]     1.19
## alpha[10]    1.19
## alpha[11]    1.14
## alpha[12]    1.10
## alpha[13]    1.15
## alpha[14]    1.15
## alpha[15]    1.10
## alpha[16]    1.09
## alpha[17]    1.13
## alpha[18]    1.16
## alpha[19]    1.14
## alpha[20]    1.15
## alpha[21]    1.14
## alpha[22]    1.23
## alpha[23]    1.04
## beta1        1.15
## beta2        1.24
## beta3        1.26
## lp__         1.46
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 00:30:16 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

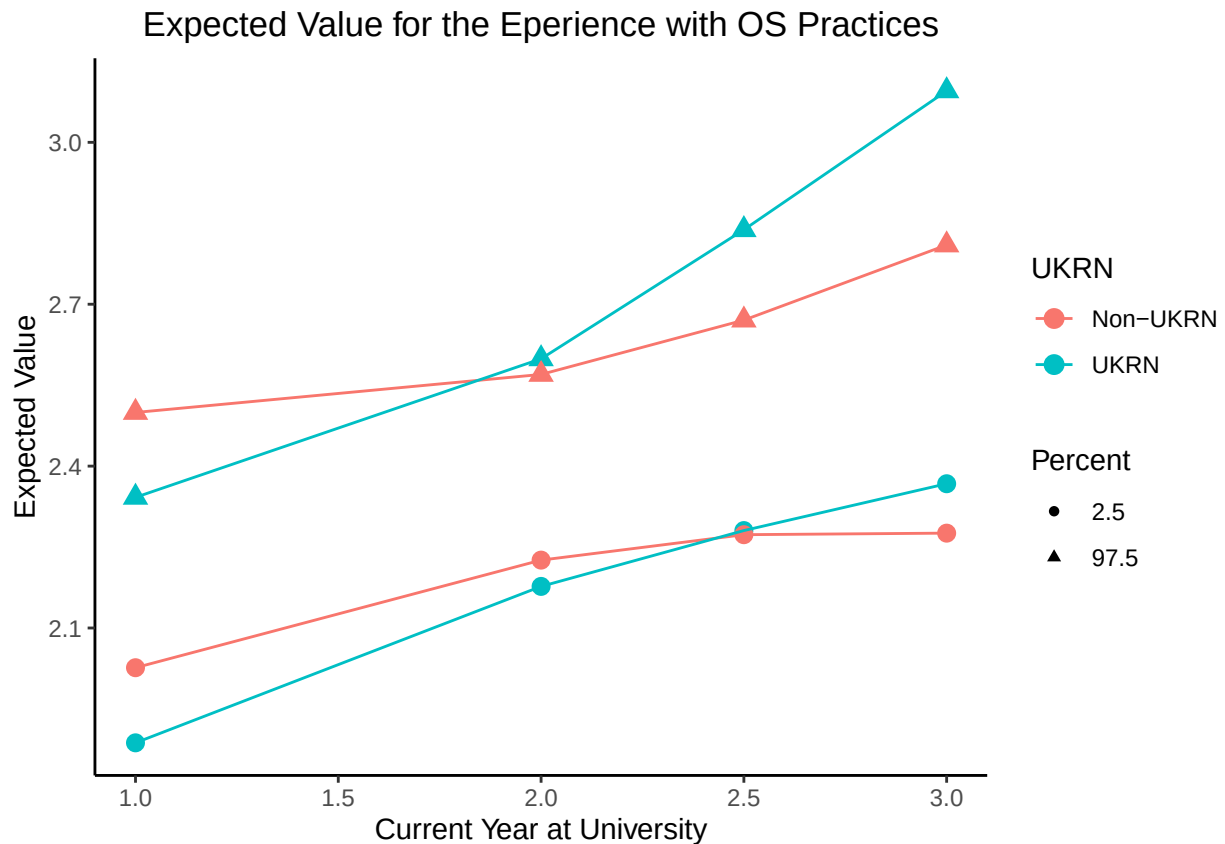
Number of participants in the analysis

Total N = 842  
 UKRN N = 435  
 Non-UKRN N = 407  
 Year 1 N = 344  
 Year 2 N = 273  
 Placement Year N = 59  
 Year 4 N = 166

| Var1 | Freq |
|------|------|
| 0    | 138  |
| 1    | 169  |
| 2    | 159  |
| 3    | 165  |
| 4    | 103  |
| 5    | 65   |
| 6    | 28   |
| 7    | 9    |
| 8    | 6    |

The output shows that all beta credibility intervals cross 0.

1207 divergent transitions. In addition: - There were 2 chains where the estimated Bayesian Fraction of Missing Information was low. - The largest R-hat is 1.02, running the chains for more iterations may help. - Effective Samples Size (ESS) is too low, indicating posterior means and medians may be unreliable. - Effective Samples Size (ESS) is too low, indicating posterior variances and tail quantiles may be unreliable.



## Data Set 2 - Linear Regression

### Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research.

The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement or third year

\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean      sd   2.5%    25%    50%    75%   97.5% n_eff Rhat
## mu           0.58     0.00   0.05   0.47   0.54   0.58   0.61   0.68  1557 1.00
## sigma        0.35     0.00   0.01   0.33   0.34   0.35   0.35   0.36  5055 1.00
## sigma_a      0.06     0.00   0.03   0.01   0.05   0.06   0.08   0.12    56 1.08
## alpha[1]     0.53     0.00   0.07   0.39   0.48   0.53   0.58   0.65   432 1.02
## alpha[2]     0.53     0.00   0.08   0.37   0.48   0.54   0.58   0.67   696 1.01
## alpha[3]     0.58     0.00   0.07   0.44   0.53   0.58   0.62   0.71  2107 1.00
## alpha[4]     0.60     0.00   0.07   0.46   0.55   0.59   0.64   0.74  2494 1.00
## alpha[5]     0.51     0.00   0.08   0.34   0.46   0.51   0.57   0.65   368 1.02
## alpha[6]     0.68     0.00   0.08   0.53   0.61   0.67   0.73   0.86   361 1.02
## alpha[7]     0.58     0.00   0.07   0.43   0.53   0.58   0.63   0.73  2515 1.00
## alpha[8]     0.61     0.00   0.07   0.48   0.57   0.61   0.66   0.75  2056 1.00
## alpha[9]     0.56     0.00   0.07   0.41   0.51   0.56   0.60   0.70  2234 1.01
## alpha[10]    0.56     0.00   0.07   0.41   0.51   0.57   0.61   0.70  2487 1.00
## alpha[11]    0.59     0.00   0.08   0.44   0.55   0.59   0.64   0.75  2942 1.00
## alpha[12]    0.52     0.00   0.08   0.36   0.47   0.53   0.58   0.66   513 1.01
## alpha[13]    0.60     0.00   0.07   0.46   0.56   0.60   0.64   0.74  2981 1.00
## alpha[14]    0.54     0.00   0.07   0.39   0.49   0.54   0.59   0.67   877 1.01
## alpha[15]    0.60     0.00   0.07   0.45   0.55   0.60   0.65   0.75  3019 1.00
## alpha[16]    0.60     0.00   0.06   0.47   0.56   0.59   0.63   0.71  2504 1.00
## alpha[17]    0.56     0.00   0.07   0.42   0.51   0.56   0.60   0.68  2274 1.00
## alpha[18]    0.55     0.00   0.06   0.43   0.51   0.56   0.59   0.67  1780 1.01
## alpha[19]    0.58     0.00   0.07   0.44   0.53   0.58   0.62   0.72  2717 1.00
## alpha[20]    0.63     0.00   0.06   0.52   0.59   0.63   0.67   0.75  1360 1.01
## alpha[21]    0.60     0.00   0.06   0.48   0.56   0.60   0.64   0.72  2512 1.00
## alpha[22]    0.62     0.00   0.06   0.50   0.58   0.61   0.66   0.74  2139 1.00
## alpha[23]    0.55     0.00   0.07   0.40   0.50   0.56   0.60   0.69  2386 1.01
## beta1       -0.04     0.00   0.08  -0.18  -0.09  -0.04   0.01   0.12  1264 1.01
## beta2        0.01     0.00   0.02  -0.04  -0.01   0.01   0.02   0.05  1686 1.00
## beta3        0.05     0.00   0.03  -0.01   0.03   0.06   0.08   0.12  2341 1.01
## lp__        520.07     2.04  10.21  506.52  514.06  518.21  522.84  553.76    25 1.21
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 00:47:24 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

### Number of participants in the analysis

Total N = 835

UKRN N = 432

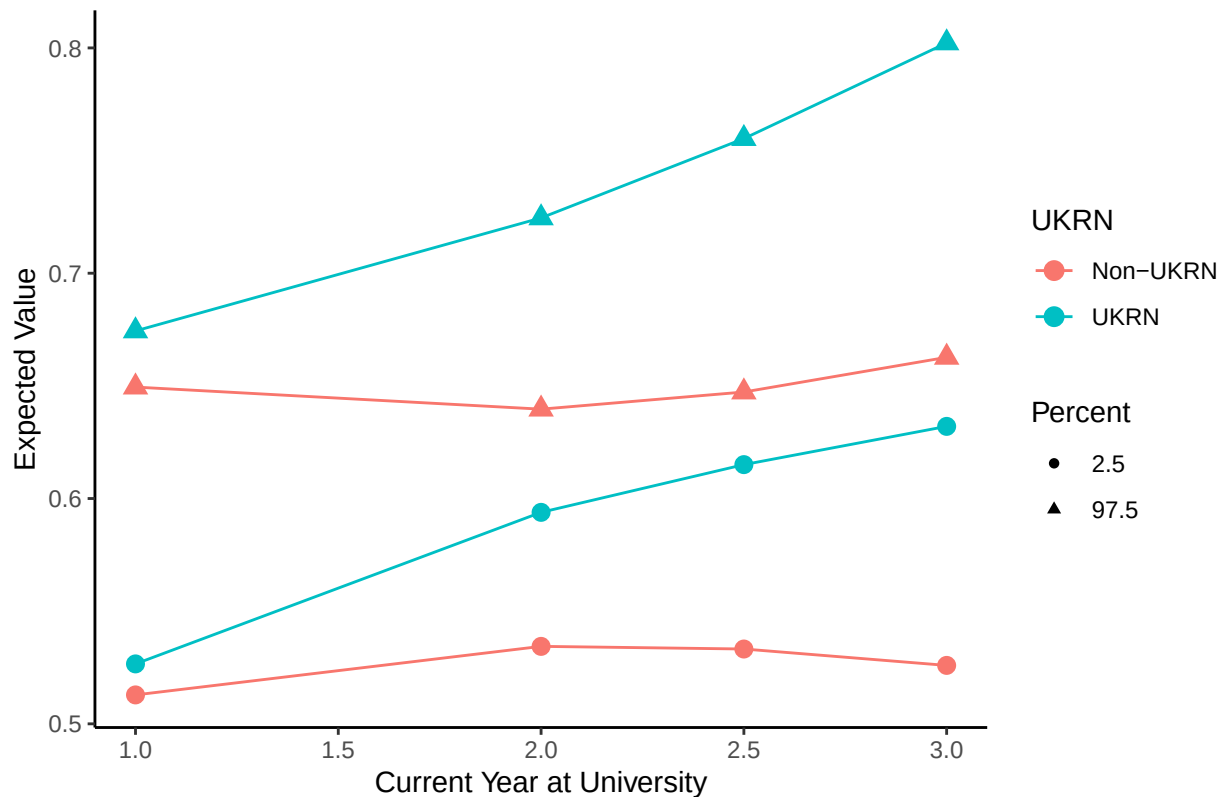
Non-UKRN N = 403

Year 1 N = 340  
Year 2 N = 270  
Placement Year N = 59  
Year 4 N = 166

The output shows that all beta credibility intervals cross 0.

234 divergent transitions.

### Expected Value for positive Conceptual Perceptions of OS Practices



### Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research.

The explainer variables are:

\* Whether the participant was at a university that was affiliated with the UKRN

\* Whether the participant was in their first, second, placement or third year

\* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean  sd  2.5%  25%  50%  75%  97.5% n_eff Rhat
## mu      0.36   0.00 0.08   0.20  0.31  0.36  0.42   0.52 2331   1
## sigma   0.48   0.00 0.01   0.46  0.47  0.48  0.49   0.51 16853  1
## sigma_a 0.14   0.00 0.04   0.08  0.12  0.14  0.17   0.23  7183  1
## alpha[1] 0.20   0.00 0.11  -0.02  0.13  0.20  0.27   0.41  2585  1
## alpha[2] 0.24   0.00 0.13  -0.01  0.16  0.24  0.33   0.48  3544  1
## alpha[3] 0.31   0.00 0.12   0.07  0.23  0.31  0.38   0.54  3076  1
```

```

## alpha[4]      0.40      0.00 0.12   0.16   0.32   0.40   0.49   0.64 3464   1
## alpha[5]      0.19      0.00 0.13  -0.09   0.11   0.20   0.28   0.45 3639   1
## alpha[6]      0.61      0.00 0.13   0.36   0.52   0.61   0.70   0.88 4110   1
## alpha[7]      0.49      0.00 0.13   0.24   0.40   0.49   0.57   0.75 4056   1
## alpha[8]      0.45      0.00 0.12   0.22   0.37   0.45   0.52   0.68 3177   1
## alpha[9]      0.31      0.00 0.12   0.08   0.23   0.32   0.40   0.55 4613   1
## alpha[10]     0.40      0.00 0.12   0.17   0.32   0.40   0.48   0.64 5208   1
## alpha[11]     0.43      0.00 0.13   0.17   0.34   0.42   0.51   0.70 5004   1
## alpha[12]     0.25      0.00 0.13  -0.01   0.17   0.25   0.34   0.50 4920   1
## alpha[13]     0.40      0.00 0.11   0.18   0.33   0.40   0.47   0.62 4849   1
## alpha[14]     0.31      0.00 0.11   0.08   0.23   0.31   0.38   0.53 4486   1
## alpha[15]     0.44      0.00 0.13   0.20   0.35   0.44   0.52   0.69 5018   1
## alpha[16]     0.37      0.00 0.09   0.18   0.30   0.37   0.43   0.55 3992   1
## alpha[17]     0.25      0.00 0.11   0.03   0.17   0.24   0.32   0.45 4662   1
## alpha[18]     0.37      0.00 0.10   0.19   0.31   0.37   0.43   0.56 4536   1
## alpha[19]     0.28      0.00 0.12   0.04   0.20   0.28   0.36   0.51 4254   1
## alpha[20]     0.44      0.00 0.09   0.26   0.38   0.44   0.50   0.61 3367   1
## alpha[21]     0.39      0.00 0.09   0.21   0.33   0.39   0.46   0.57 4075   1
## alpha[22]     0.51      0.00 0.09   0.33   0.45   0.51   0.57   0.69 3648   1
## alpha[23]     0.29      0.00 0.13   0.02   0.20   0.29   0.37   0.54 5079   1
## beta1         -0.02      0.00 0.12  -0.26  -0.11  -0.03   0.06   0.22 2992   1
## beta2         -0.02      0.00 0.03  -0.08  -0.04  -0.02   0.01   0.05 2516   1
## beta3          0.12      0.00 0.05   0.02   0.09   0.12   0.15   0.21 3997   1
## lp__          218.01      0.06 4.52 208.43 215.13 218.29 221.18 226.08 5517   1
##
## Samples were drawn using NUTS(diag_e) at Fri Feb 20 01:04:03 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

#### Number of participants in the analysis

Total N = 803

UKRN N = 421

Non-UKRN N = 382

Year 1 N = 326

Year 2 N = 261

Placement Year N = 57

Year 4 N = 159

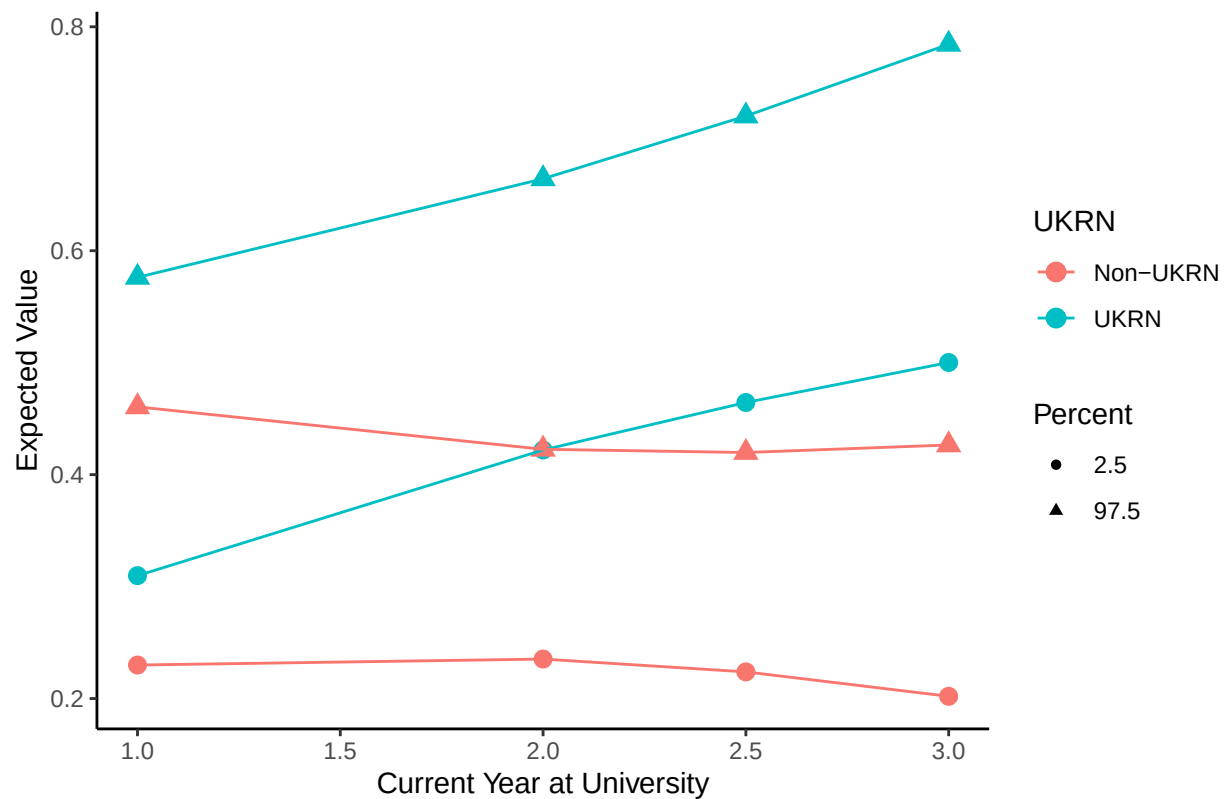
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0237474

97.5% = 0.2128806

The interpretation is that students who attend a UKRN affiliated university are more likely to have positive situational perceptions of open and reproducible practices in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

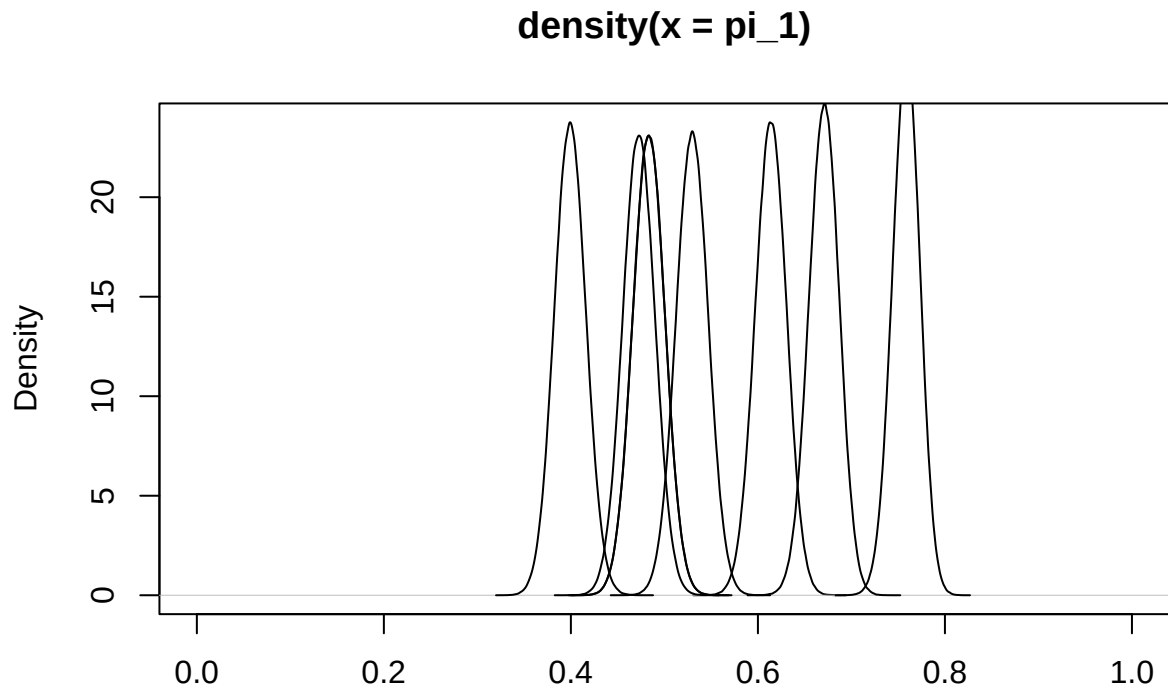
### Expected Value for positive Conceptual Perceptions of OS Practices



### Dataset 2 - Beta Bernoulli Model

For the most common category in the Awareness variable

The categories are: \* Replication \* Open Research Materials \* Open Educational Materials \* Open Access Journals \* Open Data \* Power Analysis \* Preregistration \* Preprint



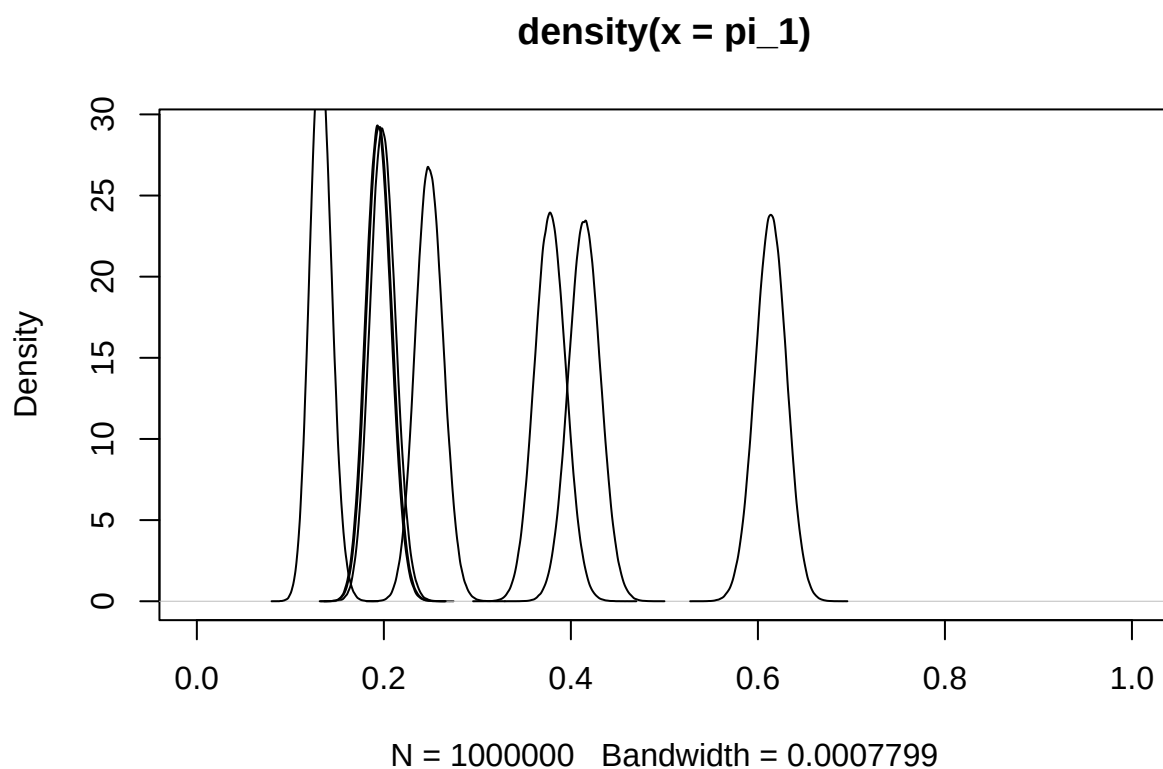
N = 1000000 Bandwidth = 0.0009511

| ##                            | Probability |
|-------------------------------|-------------|
| ## Replication                | 0.000000    |
| ## Open Research Materials    | 0.000000    |
| ## Open Educational Materials | 0.000028    |
| ## Open Access Journals       | 0.000000    |
| ## Open Data                  | 0.000000    |
| ## Power Analysis             | 0.000000    |
| ## Preregistration            | 0.000000    |
| ## Preprint                   | 0.999972    |

probs\_Awareness

**For the most common category in the Experience variable**

The categories are: \* Replication \* Open Research Materials \* Open Educational Materials \* Open Access Journals \* Open Data \* Power Analysis \* Preregistration \* Preprint



| ##                             | Probability |
|--------------------------------|-------------|
| ## Replication                 | 0           |
| ## Open Research Materials     | 0           |
| ## Open Educational\nMaterials | 1           |
| ## Open Access Journals        | 0           |
| ## Open Data                   | 0           |
| ## Power Analysis              | 0           |
| ## Preregistration             | 0           |
| ## Preprint                    | 0           |

probs\_Experience